ICV_Final_Report.docx

by Aarjan Adhikari

School of Computing, Creative Technology and Engineering

Student Name	Aarjan Adhikari
Student ID	C7466773
Module Name	Intelligent Computer Vision
Level	6
Assignment name	Coursework in Image Video Processing

I understand that to use the work and ideas of others, including generative AI output, without full acknowledgement, is academic unfair practice.

I confirm that this coursework submission is all my own, original work and that all sources, summaries, paraphrases and quotes are fully referenced as required by the LBU Academic Regulations.

DECLARATION OF GENERATIVE AI USE: (delete one of the below as appropriate)

I DID use Generative AI technology in the development, writing, or editing of this assignment. I have included a copy of the reference used, and the output generated in the Appendix section. If you have any concerns about whether you have used generative AI tools (in)correctly, please seek advice before submitting your assignment from academic staff responsible for the assessment that this submission relates to. The shaded boxes are examples to help you with completing this section.

Appendix	Generative AI Tool (e.g. ChatGPT)	How generative AI Tool was used	Reference
A	ChatGPT 3.5	Asked for some ideas on how I might evaluate the usefulness of a piece of research	OpenAI. (2023). ChatGPT (Sept 12 version) [Large language model]. https://chat.openai.com/chat
B	ChatGPT 3.5	Asked for feedback on my xxx paragraph structures	OpenAI. (2023). ChatGPT (Sept 14 version) [Large language model]. https://chat.openai.com/chat
C	Gamma	To produce slides from research and planning notes for the group presentation	OpenAI. (2023) Gamma (November 11th version). https://gamma.app/

Table of Contents

Table of Figures

Introduction

Image processing is a field of computer science that deals with understanding the nature of an image to extract information from an image or make improvement on the existing image by increasing quality, equalizing brightness, balancing colors, etc. Video processing is also mostly image processing due to video being a sequence of images called frames. When a video or image is provided, it undergoes various kinds of processes through analysis as required, such as using filtering techniques for noise removal, masking to change pixels with a set condition, or detection techniques such as edge detection, corner detection, object detection, etc.

For the ICV module final assignment, there were 6 images, 2 videos and 1 frame sequence folder with multiple frames provided. Each of the file had to be processed using the analysis and processing methods learnt throughout the module duration. The files are evaluated with 3 metrics, image histogram, PSNR, and contrast. However, PSNR metric is useful mainly for original image and the enhanced image. Therefore, PSNR is just used to evaluate the difference caused in the provided image and the processed image.

Image 1

Analysis

Looking at the image, there is a man shining a bright green torch on a pair of jeans. The whole image is tinted green due to the brightness of the light. Due to the intensity of the light, the part on the jeans shows white instead of green. So, the green tint has to be removed for the processing. Also, the brightness of the torch has to be dampened, then the brightness can be globally equalized to highlight the details of the image.

Processing

For processing, the color space is converted from RGB to HSV. This is because HSV separates color from brightness. Hence, for removing the green tint, the Hue channel can be worked on separately. Then, a mask is created where green lies in the image. Saturation and Values are also reduced from the green mask. A bilateral filter is applied to reduce the noise in the image (Tomasi and Manduchi, 1998).

Metrics

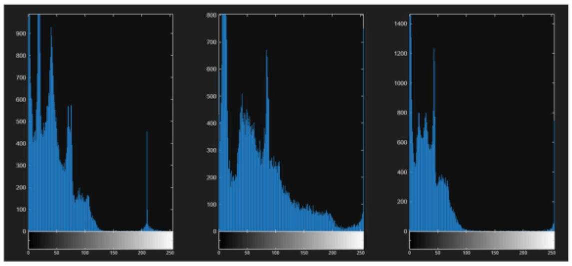

Figure i Histogram of original

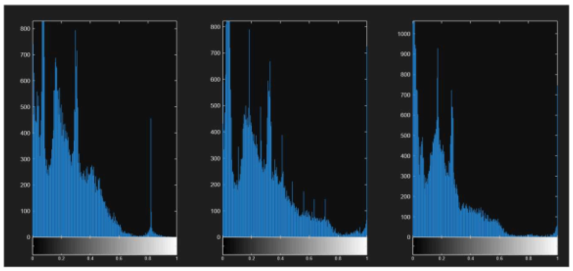

Figure ii Histogram of enhanced

Comparing the histogram, there is a shift of red a bit towards the center, blue is almost the same, and yellow is also spread towards the center.

PSNR: 11.4544

Contrast: 46.7762

Enhanced Contrast: 0.29466

Conclusion

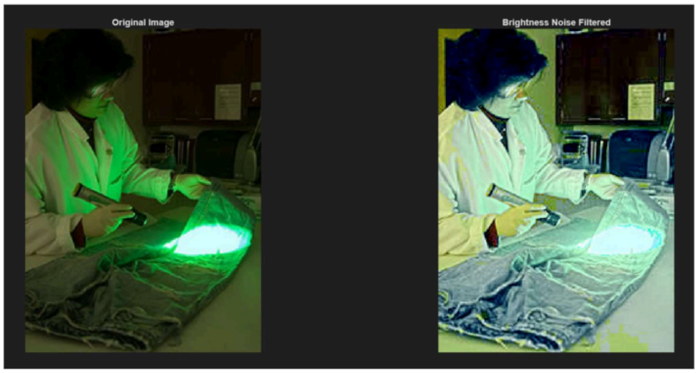

Figure iii Original vs enhanced

The Image is less saturated with green and details are shown better. There are some spots where color was lost, in the bottom right, which was caused due to the green overtaking the color information of the table which was removed.

Image 2

Analysis

In this image, a man is standing on a backyard of some kind. The man is out of focus and the background is quite dark due to the shadows. Removing the blur to reveal the person is a big challenge and out of reach of the syllabus. However, the contrast of the image can be balanced and a filter can be used to reduce the blur.

Processing

The image was converted to double to sharpen the details in the background. The image was then converted to lab for equalizing histogram of the luminescence. The image was taken back to RGB, then filtered using bilateral filtering to smoothen details.

Metrics

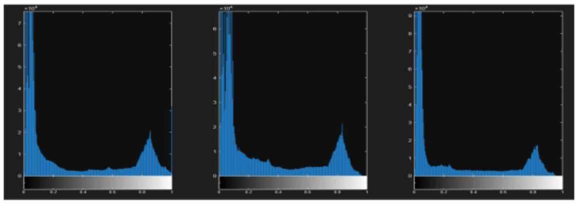

Figure iv Histogram of original

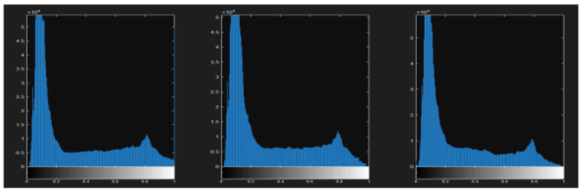

Figure v Histogram of enhanced

The histogram shows all three of the color channels being spread towards the center. There is an obvious spike at the 0 intensities for all the channels but it is fine due to the image not only having a large number of shadow affected areas, but the background itself contains a dark greenery.

PSNR: 60.4773

Contrast: 0.3027

Enhanced Contrast: 0.27142

Conclusion

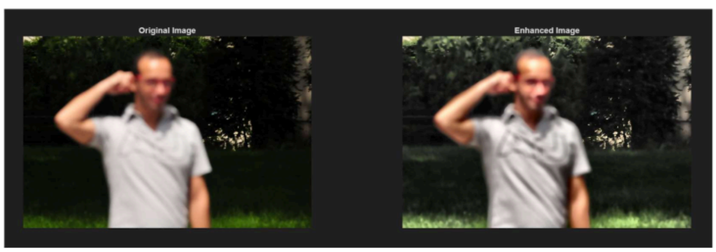

Figure vi Original vs enhanced

The image has a lighter and detailed background. While the person is not deblurred, the overall quality of the image looks cleaner. The only downside is the saturation is a bit lower than the original.

Image 3

Analysis

The image contains a car in a yard. The image is in a grayscale format with a lot of noise. Due to the image already losing too much information of the car's license plate, finding out the number of the plate is unnecessary. So, the image can be processed by removing the noise and smoothening the image. Edge detection can also be done to demonstrate edge detection in grayscale images.

Processing

The image was applied with different filters for testing. Visually, bilateral filtering worked best and was used as the initial filtered image for edge detection. A list called techs (techniques) was created and in every loop, a binary image was created using each of the edge detection techniques. Canny old was found to be the best (Sekehravani et al., 2020). A mask was created where edge was detected and of the color red. The mask was applied to the bilateral filtered image to show the edges.

Metrics

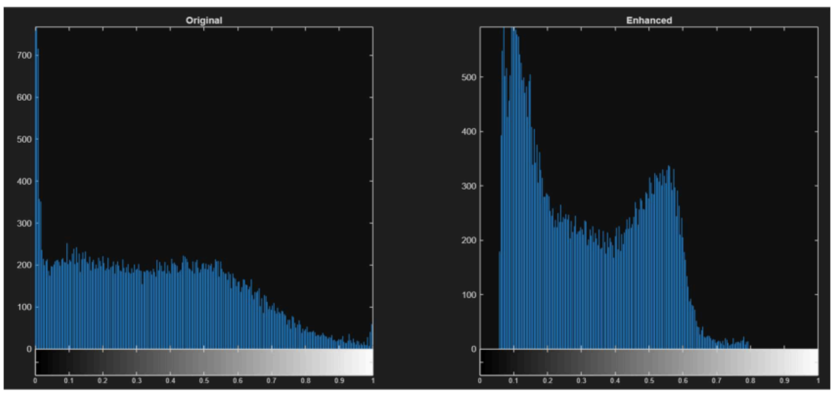

Figure vii Histogram of original vs enhanced

Compared to other images, this one has only one channel for original and enhanced image as this is a grayscale image. The histogram of the enhanced image shows a bigger spike towards the center and with more spikes in the black and almost no spike in the white intensities. This is because the noise in the original contained a lot of large white pixel chunks. The overall intensity of the image shifted low due to the denoising. However, it was not equalized again to make the histogram equal again because details were lost and the image became unnecessarily bright.

PSNR: 57.0828

Contrast: 0.24973

Enhanced Contrast: 0.18167

Conclusion

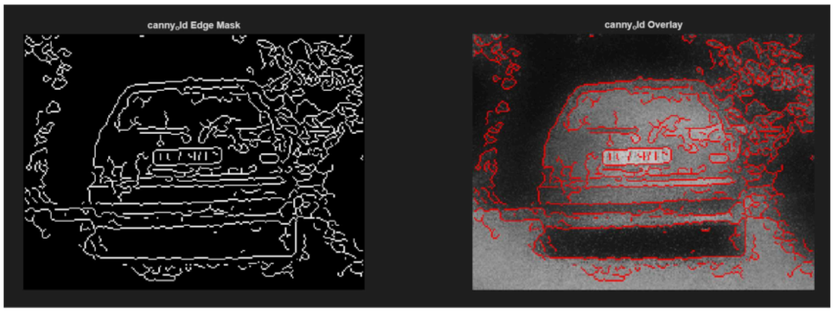

Figure viii edge detection mask vs overlay

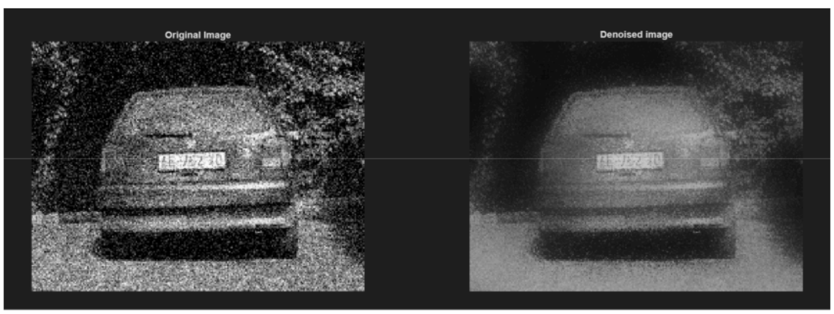

Figure ix Original vs enhanced

The edge detection worked very well and highlights the car, along with the surrounding trees. The filtered image done by bilateral filtering also makes the image very smooth with a balanced contrast.

Image 4

Analysis

In the image, a person is walking down a road at what looks like dusk or dawn. There are bright street lamps which reflect the light onto the floor, making it bright. The sky has dark clouds and where the sky meets the land, there are no clouds in the sky, causing the sun behind to light up the middle region of the image. The person is facing away from the street lights, hence appears darker. For this image, color has to be equalized and the image has to be brightened to highlight the subject better. Finally, noise can be removed to make the image cleaner.

Processing

The image was converted to lab for equalizing histogram of the luminescence. First of all, local histogram equalization (CLAHE) was done through adapthisteq function (Reza, 2004). Then, the equalized image was once again applied a global brightness equalization to limit the brightness from 0 to 0.9. This was to reduce the exposure on the image. The image was then filtered using bilateral filtering and converted to HSV. Here, the global saturation was reduced along with capping brightness to 0.7. The image was once again converted to RGB to brighten the image selectively to reduce the effect of the shadow. Then, finally, another bilateral filtering was performed to reduce the noise caused by brightening the image.

Metrics

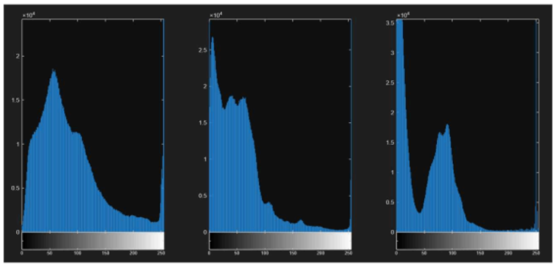

Figure x Histogram of original

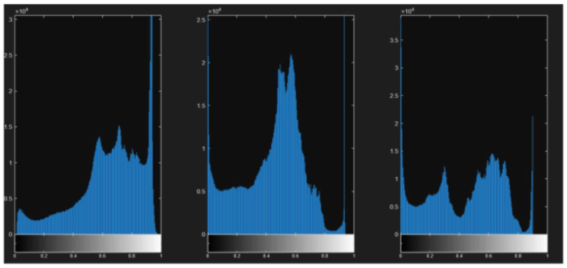

Figure xi Histogram of enhanced

The histogram comparison shows a clear difference in the spike levels. All the color channels have shifted right towards high intensities which may seem like they counteract the argument of the image by making the light sources brighter, but since the shifts are in all channels, and due to previous brightness equalization, the image is perfectly brightened up.

PSNR: 8.9417

Contrast: 56.1777

Enhanced Contrast: 0.21694

Conclusion

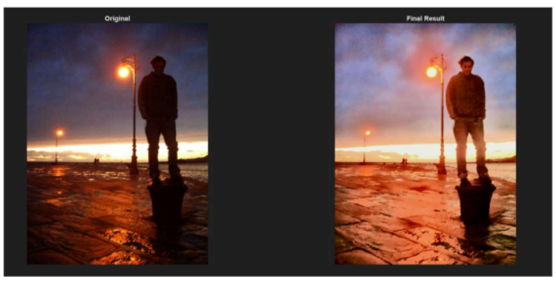

Figure xii Original vs enhanced

The image now is balanced in contrast and the man is more visible in the dark. The brightness on the floor is distributed and the clouds on the sky appear lighter. The transition between the sky and the land are also smoother.

Image 5

Analysis

The image consists of a person in front of a building. The person is walking perpendicularly away from the building, and the camera is facing the building at about 45 degrees. The overall quality of the image is very good but the problem is that the person appears blurry, caused by motion blur. While motion blur filter could be applied, due to the unavailability of the person's original face, utilizing the good picture quality to showcase edge detection on the buildings and surroundings is better for analysis.

Processing

Due to the quality already looking good in the original image, there were no filters or color correction methods applied here, instead only edge detection was done. First, during the testing phase, the same method as done in Image 3 where each method available for testing was done. However, unlike in Image 3, Roberts edge detection was used instead of Canny old or others due to others either extracting too many edges or too less. The binary image of the edge was also applied with imdilate to highlight edges better, and bwareaopen was used to finally connect edges that were not detected properly.

Metrics

Due to only edge detection being applied here, there were no metrics calculation.

Conclusion

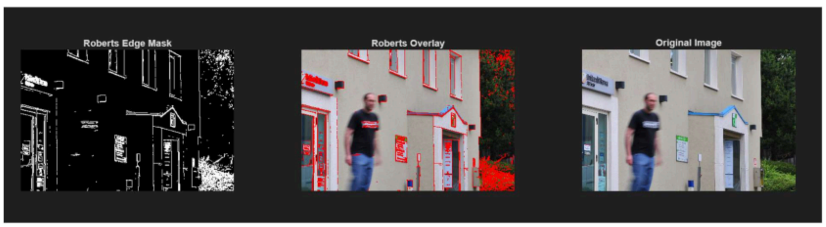

Figure xiii edge mask vs enhanced vs original

The edge detection cleanly detected all edges. As seen in the image, the person is not detected, but it was intentional as the person was not the primary target. As discussed during processing, Roberts was chosen as it didn't consider too many edges and the person was one of the edges not considered while processing. Others like canny, log, etc. considered the person as edges.

Image 6

Analysis

The image consists of some soldiers kneeling on the ground. The image has a “hay” like color tint, which makes the image look very vintage. Nothing too concerning other than the tint can be observed. Hence, to process this, removing the tint and smoothening the image if any blur or noise occurs during processing is enough.

Processing

Due to the tint of the image, it is best to convert to the LAB color space. The advantage of LAB is, the luminance portion of the image is separated from the color information. The mean values of the a and b chromatic channels were subtracted to reduce global color bias under the Gray-World assumption. This approach helps restore more natural color representation while preserving luminance information (Kim and Lee, 2023). Then, simply a bilateral filter can be applied to remove the noise that occurred. The color components, after being subtracted the mean, is converted back to RGB. Then, the RGB is converted to HSV for increasing the saturation . After once again converting to RGB, the image is then filtered using bilateral filtering.

Metrics

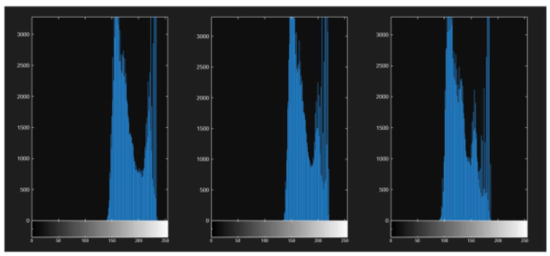

Figure xiv Histogram of original

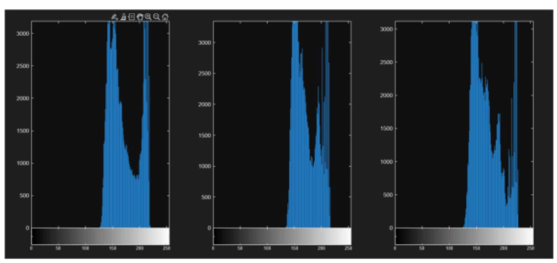

Figure xv Histogram of enhanced

The histogram of this image may not be very clear but a small change can be observed. Red channel is shifted slightly to the left as it contributes to the yellow tint. Blue is shifted more to the right as the final image is very light colored. So, blue has to be increased to get the equal intensity mix of red, green and blue to get white.

PSNR: 20.807

Contrast: 25.3684

Enhanced Contrast: 24.1995

Conclusion

Figure xvi Original vs enhanced

The image shows the yellow tint removed from the image successfully. The image is also saturated further to not let the yellow removal affect the quality. Also the deblurring makes the image look smoother.

Video 1

Analysis

The video is 3 seconds long with a frame rate of 7 fps. It consists of a car trying to forward park while facing away from the camera. The license plate number is already clearly visible, “EX 764P2” so quality improvement seems unnecessary. So, this video looks like a good opportunity to highlight edge detection in a video.

Processing

For the edge detection, first of all, each frame of the video has to be converted to grayscale. Bilateral filtering is used to slightly smoothen the image. Then, using canny edge detection, edges are calculated. bwareaopen is used to connect disconnected edges. Just like previous images, a red border has been applied to the image. The whole process is applied to every image and the video file is also updated with each edge detected frame.

Conclusion

Figure xvii Sample frame from the video highlighting edge detection

The snapshot of the video shows that the edges are being detected properly. Also the bilateral filtering has made the video a bit smoother than the original.

Video 2

Analysis

The video is 4 seconds long with a frame rate of 15 fps. The video consists of a car being recorded in the dark. However, the license plate is bright and even though it is hard to see, “CB 9169H” can be seen written on the plate. Since the video contains almost no background information that seems necessary to extract, sharpening the quality of the license plate, even though the number can be seen, is the better approach for the processing of this video.

Processing

For this video, each frame was processed in LAB color space. First of all, CLAHE was done through adapthisteq for balancing the luminescence. Then, the frames were converted back to RGB in order to apply bilateral filtering to smoothen out the frame, which was then sharpened using imsharpen. Due to slight calculation errors, the values escaped the normal range of 0 and 1. Hence, with the combination of max and min functions, the values were clamped back to 0 and 1.

Conclusion

Figure xviii Sample frame from the video showing clearer image

The license plate area has been brighter, and the license plate numbers look like they are anti-aliased compared to the original. The overall contrast of the image has risen due to brightness equalization.

Frame Sequence

Analysis

There are 240 sets of images numbered vespa001…vespa240.jpg. The images are frames of a video supposed to be reconstructed first before analysis. Hence, first of all, a separate file has to be created to create a video from the frame sequence. Constructing the video, 24 fps seems like the perfect rate. After the video is constructed, the video shows a man riding a scooter at night. The license plate reads “TS-6 1331” or similar and can be observed slightly during some specific frames. The video has lots of camera shake and is of a very poor quality. For processing, the contrast can be equalized, image can be denoised, and edge detection can also be applied.

Processing

First of all, the frame sequences were read one at a time from the directory and were written as a video file called “video3.avi”. Then, rest of the reading and processing were done on another file. For each frame of the video, first of all, a gamma correction was done by multiplying each element with 0.6. Then, the brightness was adjusted in the image using imadjust. For each frame, bilateral filtering was used to reduce noise. Then, canny edge detection was used to detect edges along with a bwareaopen function to close edges and imdilate to increase the width of the edges. Next, a red border overlay was created to display the edges.

Conclusion

Figure xix Sample frame from the video showing enhanced image

The edge detection did not work properly in the image. However, other things have been definitely improved including the contrast equalization, brightness equalization and the denoising of the frames. Considering the amount of information loss on the video, the video processing can be considered a success.

Conclusion

The project showcases different kinds of scenarios and challenges that can occur during image processing and video processing. There are different ways to approach any scenarios but also there are many common techniques that can be applied for general case solutions. As each image or videos were processed, it also became very clear how each scenario should be approached and how different functions can be used together and also how a change can affect the entire processing pipeline.

The report concludes the ICV module which has taught various kinds of approach for signal processing both theoretically and practically. While MATLAB did provide functions for making the tasks simple, the clarification of the steps required were perfect from the lecture slides for better understanding.

Bibliography

Acharjya, P.P., Ghoshal, D. and Das, R. (2020) 'Detection of edges in digital images using edge detection operators', *International Journal of Computer Science Engineering*, 9(1), pp. 1–6. Available at: https://www.researchgate.net/publication/356379177 (Accessed: 3 May 2026).

Kim, J. and Lee, S. (2023) ‘Saturation-Based Airlight Color Restoration of Hazy Images’, *Applied Sciences*, 13(22), p. 12186. Available at: https://www.mdpi.com/2076-3417/13/22/12186 (Accessed: 4 May 2026).

Reza, A.M. (2004) 'Realization of the Contrast Limited Adaptive Histogram Equalization (CLAHE) for Real-Time Image Enhancement', *Journal of VLSI Signal Processing Systems for Signal, Image, and Video Technology*, 38(1), pp. 35–44. Available at: https://doi.org/10.1023/B:VLSI.0000028532.53893.82 (Accessed: 14 May 2026).

Sekehravani, E.A., Babulak, E. and Masoodi, M. (2020) 'Implementing canny edge detection algorithm for noisy image', *Bulletin of Electrical Engineering and Informatics*, 9(4), pp. 1404–1410. Available at: https://doi.org/10.11591/eei.v9i4.1837 (Accessed: 7 May 2026).

Tomasi, C. and Manduchi, R. (1998) 'Bilateral filtering for gray and color images', in *Proceedings of the Sixth International Conference on Computer Vision (ICCV)*. Bombay, India: IEEE, pp. 839–846. Available at: https://doi.org/10.1109/ICCV.1998.710815 (Accessed: 3 May 2026).

ICV_Final_Report.docx

ORIGINALITY REPORT

13%	9%	4%	12%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Leeds Beckett University Student Paper	7%
2	dl.ucsc.cmb.ac.lk Internet Source	1%
3	Submitted to University of Cape Town Student Paper	1%
4	iieta.org Internet Source	1%
5	www.mdpi.com Internet Source	1%
6	www.webology.org Internet Source	<1%
7	Submitted to The British College Student Paper	<1%
8	Submitted to Queen Mary and Westfield College Student Paper	<1%
9	www.shotuk.org Internet Source	<1%
10	ijfans.org Internet Source	<1%
11	repo.lib.duth.gr Internet Source	<1%

Exclude quotes Off

Exclude matches Off

Exclude bibliography Off